Unnumbered continuation: actual filled carriers and finite controlled expansions. Retain the original data and pointed group $\mathscr{F}$ of (CFT.1). Let P be a compact metric space with specified point u_0, and let $u \mapsto h_u \in \mathscr{F}$ be continuous in the original closed-disk map-and-inverse topology, with $h_{u_0} = \mathrm{id}$. Every map below uses this given family and its specified lifts.

Fix a homeomorphism $\iota : \overline{D} \to \overline{\mathbb{D}}$, and write

$$e(z,w) = |\iota z - \iota w|, \qquad e_Q(z,w) = \sup_{q \in Q} e(qz, qw). \tag{FCE.1}$$

This is a fixed coordinate on the original closed disk. It is not a change of the operator action. Define the two comparison functions for every $a \geq 0$ by

$$\omega_{d,e}(a) = \max_{d(z,w) \leq a} e(z,w), \qquad \omega_{e,d}(a) = \max_{e(z,w) \leq a} d(z,w). \tag{FCE.2}$$

The maxima exist by compactness. Both functions are nondecreasing, vanish at zero and tend to zero at zero: a contrary sequence has a convergent subsequence of pairs with equal limit points. The identity map and its inverse therefore obey $e_Q \leq \omega_{d,e} \circ d_Q$ and $d_Q \leq \omega_{e,d} \circ e_Q$. For the sums D_e, D_d of the original uniform map and inverse distances, the same pointwise estimates give $D_e \leq 2\omega_{d,e}(D_d)$ and $D_d \leq 2\omega_{e,d}(D_e)$. These are both topology comparisons, on the original maps with the original labels.

Put

$$a^+(y) = \max_{u \in P} e_Q(h_u z, z), \qquad a^-(y) = \max_{u \in P} e_Q(h_u^{-1} z, z), \quad y = \pi z. \tag{FCE.3}$$

Equivariance makes these independent of the representative z. The proof of (CFT.12)–(CFT.15), applied to e using (FCE.2), shows that $a^+, a^- \in C_0(N)$. In particular both of the actual displacements have a common continuous positive C_0 majorant, for example $a^+ + a^- + \tau$, where τ is the explicit function constructed in (CFT.14).

Filling an actual compact connected set. For a nonempty compact connected $A \subset O$, let $\mathrm{Fill}(A)$ be A together with all components of $O \setminus A$ whose closures in O are compact. In the coordinate ι, choose $r < 1$ with $\iota A \subset \{|z| < r\}$. The connected annulus $r \leq |z| < 1$ lies in a single component U_A of $O \setminus A$. Every other component has closure contained in the compact disk $|z| \leq r$; hence $\mathrm{Fill}(A) = O \setminus U_A$ is compact. It is connected because each filled component has boundary in A, and that boundary is nonempty. For the nonempty-boundary assertion, a component with empty boundary would be open and closed in the connected space O while missing the nonempty set A.

Every point outside the Euclidean convex hull of ιA has a separating straight line and a ray to the unit circle which misses that convex hull. It therefore lies in U_A. Consequently

$$\iota\,\mathrm{Fill}(A) \subset \mathrm{conv}(\iota A), \qquad \mathrm{diam}_e\,\mathrm{Fill}(A) = \mathrm{diam}_e\,A. \tag{FCE.4}$$

To verify the diameter equality, two finite convex combinations $\sum_i \alpha_i v_i$ and $\sum_j \beta_j w_j$ have difference $\sum_{i,j} \alpha_i \beta_j (v_i - w_j)$, of norm at most $\mathrm{diam}(\iota A)$. Thus the convex hull has this same diameter; the reverse inequality uses $A \subset \mathrm{Fill}(A)$.

A homeomorphism $q : O \to O$ maps complement components bijectively and maps compact closures to compact closures; its inverse proves the reverse implication. Thus the exact set map and its inverse give

$$q\,\mathrm{Fill}(A) = \mathrm{Fill}(qA), \qquad \mathrm{diam}_{e_Q}\,\mathrm{Fill}(A) = \sup_{q \in Q} \mathrm{diam}_e(qA) = \mathrm{diam}_{e_Q}\,A. \tag{FCE.5}$$

This uses no extension of q to the circle.

Let A now be a compact arc with endpoint w fixed by every h_u. The compact connected set $A_P = A \cup \bigcup_{u\in P} h_u(A)$ has that common point. Joint continuity on $P \times A$ proves compactness even when P is not connected. Equivariance gives the exact formula

$$q\,\mathrm{Fill}(A_P) = \mathrm{Fill}\left(qA \cup \bigcup_{u\in P} h_u(qA)\right), \qquad \mathrm{diam}_{e_Q}\,\mathrm{Fill}(A_P) \le \mathrm{diam}_{e_Q}\,A + 2\sup_{y\in\pi A} a^+(y). \tag{FCE.6}$$

Indeed any two points of the union on the right are images of two points of qA, or are themselves such points. The triangle inequality through those two original points gives the displayed bound, uniformly in q, and (FCE.4)–(FCE.5) transfer it to the filled set. In particular the estimate concerns every actual translated carrier in the original disk.

An explicit local expansion of a common fixed region. Choose an embedding $\zeta : [-1,1]^2 \to O$ whose closed image D has pairwise disjoint Q-translates and with $x \notin QD$. Such disks exist in any sufficiently small covering chart away from Qx. Let $K \subset O$ be closed and Q-invariant, fixed pointwise by every h_u, and suppose its intersection with the chosen disk is exactly $\zeta\{(s,v) : v \le 0\}$. These are the local fixed-set data for this operation; no such chart for an arbitrary fixed set is inferred.

Choose $0 < a_0 < 1$ and a continuous $\theta : [-1,1] \to [0,1]$ equal to zero near the two endpoints. For $0 \le a \le a_0$ set

$$F_a(v) = \begin{cases} a + (1+a)v, & -1 \le v \le 0, \\ a + (1-a)v, & 0 \le v \le 1, \end{cases} \qquad F_a^{-1}(w) = \begin{cases} (w-a)/(1+a), & -1 \le w \le a, \\ (w-a)/(1-a), & a \le w \le 1. \end{cases} \tag{FCE.7}$$

Both composites are the identity on their two stated pieces. The formulas agree at their breakpoints, and $1 - a \ge 1 - a_0 > 0$. Thus maps and inverses vary jointly continuously with a. For $t \in [0,1]$ define

$$\tau_t(q\zeta(s,v)) = q\zeta\big(s, F_{ta_0\theta(s)}(v)\big), \quad q \in Q, \qquad \tau_t(z) = z \quad (z \notin QD). \tag{FCE.8}$$

The disjoint translates make this definition unambiguous. It fixes every disk boundary; properness of Q makes the translates of the compact disk locally finite in O. The inverse is the same formula with F_a^{-1}, pasted along the same boundaries. These facts prove joint interior continuity of both maps and inverses. Their quotient support lies in the one compact set πD, so C.81 gives joint extension by the identity to the original circle. They commute with every original operator and fix x.

The exact image of K in qD is now

$$\tau_t K \cap qD = q\zeta\{(s,v) : v \le ta_0\theta(s)\}, \qquad K \subset \tau_t K, \qquad H_{u,t} = \tau_t h_u \tau_t^{-1}. \tag{FCE.9}$$

Outside QD, $\tau_t K$ equals K. The inclusion follows from $ta_0\theta(s) \ge 0$. Each $H_{u,t}$ fixes $\tau_t K$ pointwise because $H_{u,t}(\tau_t k) = \tau_t h_u k = \tau_t k$. It lies in $\mathscr{F}$, has inverse $\tau_t h_u^{-1}\tau_t^{-1}$, starts at h_u, and is the identity for $u = u_0$. Thus it is an actual based deformation which enlarges this common fixed region and retains K.

Here the topology is the original one. On a compact disk, uniform convergence $f_n \to f$, $g_n \to g$ gives uniform convergence of $f_n g_n$ to fg: the difference is bounded by $\sup e(f_n, f)$ plus the modulus of uniform continuity of f applied to $\sup e(g_n, g)$. Apply this also to $g_n^{-1} f_n^{-1}$ for the inverse maps. Therefore composition and inversion are continuous in the stated map-and-inverse metric. This proves continuity of $(u,t) \mapsto H_{u,t}$; (FCE.2) gives both comparisons with the original metric d.

Two actual disk families and quantitative bounds. The same estimates apply to every jointly continuous boundary-fixed isotopy on D with its equivariant paste τ_t, whether or not it is the

expansion (FCE.8). Write $S = QD$ and

$$s_D = \operatorname{diam}_{e_Q} D, \quad m_u^+ = \operatorname{diam}_{e_Q} h_u(D), \quad m_u^- = \operatorname{diam}_{e_Q} h_u^{-1}(D), \quad \alpha_D^\pm = \sup_{y\in\pi D} a^\pm(y). \tag{FCE.10}$$

The triangle inequality through two points of qD gives $m_u^\pm \le s_D + 2\alpha_D^\pm$ for every u. The exact left correction and its inverse are

$$A_{u,t} = H_{u,t}h_u^{-1} = \tau_t\,(h_u\tau_t^{-1}h_u^{-1}), \qquad A_{u,t}^{-1} = h_u\tau_t h_u^{-1}\tau_t^{-1}. \tag{FCE.11}$$

The first factor is pasted on the disks qD. The second is pasted on the disks $h_u(qD) = qh_u(D)$, which are pairwise disjoint because h_u is injective. The individual maps are exactly the displayed conjugates; the two families may overlap with one another. This factorization retains all original deck labels without choosing any new point tracks.

Every displacement of τ_t or τ_t^{-1} in the metric e is at most s_D. Every displacement of $h_u\tau_t^{\pm1}h_u^{-1}$ is at most m_u^+. The triangle inequality for the two factors in (FCE.11), and then the same argument with h_u^{-1} in place of h_u, proves

$$\begin{aligned} \sup_{z\in\overline{D}} e(H_{u,t}z, h_u z) &\le s_D + m_u^+, \\ \sup_{z\in\overline{D}} e(H_{u,t}^{-1}z, h_u^{-1}z) &\le s_D + m_u^-, \\ D_e(H_{u,t}, h_u) &\le 2s_D + m_u^+ + m_u^- \\ &\le 4s_D + 2\alpha_D^+ + 2\alpha_D^-. \end{aligned} \tag{FCE.12}$$

In the original d metric the first two bounds instead give $D_d(H_{u,t}, h_u) \le \omega_{e,d}(s_D+m_u^+)+\omega_{e,d}(s_D+m_u^-)$. Thus both maps and inverses are controlled by the actual common displacement envelope on this carrier.

The exact input regions on which a change can occur are

$$\begin{aligned} \{z : H_{u,t}z \ne h_u z\} &\subset S \cup h_u^{-1}S, \\ \{z : H_{u,t}^{-1}z \ne h_u^{-1}z\} &\subset S \cup h_u S, \\ \operatorname{supp}_O(A_{u,t}) &\subset S \cup h_u S. \end{aligned} \tag{FCE.13}$$

For example outside $S \cup h_u^{-1}S$ the first τ_t^{-1} fixes the input and the last τ_t fixes its h_u image. The inverse formula proves the second inclusion; the two factors prove the third. Here S is closed in O by local finiteness of the closed disks, and h_uS is closed by the homeomorphism h_u. Thus taking closure in O preserves the third inclusion. All these quotient carriers lie in compact sets uniformly over $u \in P$, because πD is compact and the given quotient maps and inverses evaluate continuously on it.

An actual small-displacement family requiring more than one such disk paste. For this example take the original disk to be the closed unit disk and $d = e$ to be Euclidean distance. Use the exact radial map $j : \mathbb{R}^2 \to \mathbb{D}$, $j(z) = z/(1+|z|)$, with inverse $j^{-1}(w) = w/(1-|w|)$, and let $q = jT_{(1,0)}j^{-1}$. The group $Q = \langle q\rangle$ is a free proper closed discrete action; conjugation by j verifies each property from integer translations. All plane translations conjugated by j extend by the identity on the original circle, commute with q, and act transitively. To check that extension directly, a bounded translation vector changes $j(z)$ by a quantity tending uniformly to zero as $|z| \to \infty$, by the estimate below. Thus this is an actual C.81 instance.

Put $\beta(v) = \max(0, 1-4|v|)$. For $n \ge 1$ and $0 \le s \le 1$, the actual maps and inverses are

$$F_{n,s}(a,y) = (a+s\beta(y-n), y), \qquad F_{n,s}^{-1}(a,y) = (a-s\beta(y-n), y), \qquad H_{n,s} = jF_{n,s}j^{-1}. \tag{FCE.14}$$

They commute with q and fix $x = j(0,0)$. They extend by the identity to the circle. Indeed, direct subtraction gives, for $v, w \in \mathbb{R}^2$, $|j(v)-j(w)| \le 2|v-w|/(1+\min(|v|,|w|))$. On the

support of (FCE.14) both points have $|y| \geq n - 1/4$ and their horizontal difference is at most 1. The estimate applies equally to the inverse and proves

$$\sup e(H_{n,s}z, z) \leq \frac{2}{n + 3/4}, \qquad D_e(H_{n,s}, \mathrm{id}) \leq \frac{4}{n + 3/4}. \tag{FCE.15}$$

Under $\pi j(a, y) = (a \bmod \mathbb{Z}, y)$ the support is in the compact strip $(\mathbb{R}/\mathbb{Z}) \times [n - 1/4, n + 1/4]$.

Let $I_n = [2^{-n-1}, 2^{-n}]$, and let s_n be the piecewise linear function on I_n which is zero at its two endpoints and one at its midpoint t_n. On $[0, 1]$ set $H_t = H_{n,s_n(t)}$ for $t \in I_n$, and $H_t = \mathrm{id}$ elsewhere. Both formulas agree at every shared endpoint. Formula (FCE.15) proves continuity at the accumulation point 0 in the original map-and-inverse metric. Identifying 0 and 1 gives an actual based loop in $\mathscr{F}$. The explicit based nullhomotopy replaces $s_n(t)$ by $(1 - r)s_n(t)$, $0 \leq r \leq 1$; the same uniform estimate proves joint continuity at 0. The family has a common C_0 displacement tail: outside $(\mathbb{R}/\mathbb{Z}) \times [-1, N + 1]$ only strips with $n > N$ contribute, and their bound is $2/(N + 7/4)$, for maps and inverses alike.

At the midpoint, for $v_n = j(0, n)$ one has the literal deck-step equality $H_{t_n}(v_n) = qv_n \neq v_n$. A closed Jordan disk containing both points in its interior meets its q-translate at qv_n. More generally, a map pasted into a Q-invariant family of pairwise disjoint disk interiors and fixing every disk boundary preserves each disk. If its left composition with H_{t_n} were to fix v_n, it would have to send qv_n to v_n inside one such disk. That disk and its q-translate would both contain qv_n in their interiors. They are distinct: otherwise all $q^k v_n$ would lie in one compact disk in O, contrary to properness of this integer-translation action. This contradicts the stated disjointness. The corresponding right correction would send v_n to $q^{-1}v_n$ and has the same contradiction. Thus one disk-paste stage of this specified form cannot make that correction, even for this continuous based family with the explicit nullhomotopy above.

Formulas (FCE.11)–(FCE.13) provide two separately disjoint actual disk families and allow overlap between the stages. Formulas (FCE.4)–(FCE.6) control the filled size of actual arcs and their given images, but do not assert pairwise disjointness of those filled sets. This note proves the finite operations and their exact original-topology bounds. It constructs no infinite sequence of compatible fixed-set expansions across all edges or essential cycles, and makes no full noncompact contraction assertion. The exact infinite-evaluation obstruction in C.83 is unchanged.